

Does the breakdown radius saturate at 1?

No — it equals a structural quantity, and that quantity is 1 in the graphs we had been drawing

	Result
The radius does not saturate	On components of size 2-12 it equals the separation — the distance inside the undirected component from X to the nearest member of the adjustment set — exactly, in 792 of 792 instances . The fraction at $r = 1$ is 100% at separation 1 and 0% at every separation ≥ 2 .
One formula covers everything	$r_val = \min(s, K_G0)$ in 1,572 of 1,572 instances, 100.00%, both coverage levels. It names both binding constraints: how far the adjustment set sits from the treatment, and how much the analyst claimed to know.
And session 1's ~80% is real, structural, and explained	The five random generators replicate it (79.6% at $r = 1$, against 81.0% and 78.6%). Why: 85.8% of measurable natural separations are 1 , the maximum is 4, and in a further 17.1% no member of the adjustment set is in the treatment's component at all.
None of it was definitional	Theorem 14 <i>proves</i> the optimal adjustment set is disjoint from $de(X)$, and on such sets the two criteria coincide. 0 of 5,304 radii move. Every prior result using $Z = O(G0)$ stands unchanged.
O* is still never beaten	0 across 235,534 GAC-admissible candidate sets this session, of which 92,683 never existed under back-door — on top of session 1's 249,732 instances.
Component size does not matter; separation does	At fixed separation the median radius is identical across every component size from 2 to 12. The pre-registered counter-mechanism — that larger components admit more extensions per release and so lower the radius — has no visible effect at all.

Session 5. Continues the four prior reports, none of which is modified. Branch **experiments/synth_graphs_and_heuristics**; nothing committed to main. Pre-registration written before the first run and unedited: *results/axisa2/preregistration.md*. Two of its predictions — mine — failed, and both are reported at equal prominence.

1. The definitions were not the problem

The repository's oracle was Pearl's **back-door** criterion, which is sufficient but not necessary for adjustment, so it counts as failures some graphs where Z is genuinely fine — biasing radii downward. The generalized adjustment criterion (GAC) is sound *and* complete. Session 4 had reconciled the two *toward* back-door to keep four sessions of results comparable; this session reverses that and makes the GAC the definition at both layers, keeping the back-door implementations unmodified as second oracles so nothing prior becomes unreproducible.

check	scope	cases	disagreements
Layer 1 (DAG) vs back-door	all 29,824 labelled DAGs on 4 and 5 nodes	4,711,024	227,224 — all classified , 0 unexplainable
Layer 2 vs the semantic anchor	GAC-valid in every DAG extension, all CPDAGs $n \leq 4$	74,568	0
— of which non-amenable		28,464	0

Every one of the 227,224 Layer-1 disagreements is the same thing: Z contains a descendant of X that is *off* the causal route to Y . Back-door bans it, the GAC allows it, and the GAC is right. back-door-valid $\implies$ GAC-valid with **0 implication failures**. Performance was specified up front this time — session 4's criterion came back exhaustively verified and exponential because only correctness was asked for: worst case **1.07 ms per call at $n = 20$** against a 5 ms bar.

1.1 The result that decides the phase, with a proof

Theorem 14. For the optimal adjustment set $O = \text{pa}(\text{cn}(X,Y)) \setminus (\text{cn}(X,Y) \cup \{X\})$, back-door validity and GAC validity are the same predicate, in every graph.

Proof. O is disjoint from $\text{de}(X)$: every element is a parent of a node in $\text{cn}(X,Y)$ and is itself excluded from $\text{cn}(X,Y) \cup \{X\}$, so none lies on or below a causal route from X . For any Z with $Z \cap \text{de}(X) = \emptyset$ the two criteria's first conditions both hold, and their second conditions coincide. ■

Verified twice, by different routes. The criterion sweep found 0 disagreements in 8,154 evaluations (3,360 both-valid, 4,794 both-invalid, so not vacuous). Independently, I recomputed radii by full BFS over the corrected space under each definition:

	instances	radii identical	of those with back-door $r = 1$, GAC strictly larger
$Z = O(G_0)$	5,304	100.00%	0 of 2,790 (0.00%)
Z arbitrary	16,926	70.86%	1,308 of 6,588 (19.85%)

The headline of the phase: of the instances with $r = 1$ under back-door, the fraction with a strictly larger GAC radius is **0%**. Where the definition *does* matter is arbitrary Z , and by a lot — which is exactly why the frontier had to be re-run rather than carried over. The pre-registered invariant $r_{\text{val}}(\text{GAC}) \geq r_{\text{val}}(\text{back-door})$ was asserted per instance throughout: **0 violations** in 22,230 instances, 0 in the census, 0 in the frontier, 0 in the random spot-check.

2. A generator that reaches the regime

Existing generators cannot produce the regime the question lives in. In session 4's 256-instance envelope the largest undirected component at $n = 24$ was **six vertices**, with 159 of 256 instances at 2–4. Dense Erdős-Rényi CPDAGs produce *more compelled edges*, not bigger chain components — **bigger n is the wrong lever**. A three-vertex component cannot exhibit a radius of 3 whatever n is.

The generator builds the CPDAG directly instead of sampling for it. A chordal component is grown by clique-attachment, so every parent set is a clique and no v -structure forms inside it; the first $s+1$ vertices form a spine $\text{anchor} \rightarrow \dots \rightarrow X$, and attaching to a clique cannot create a shortcut, so $d(X, \text{anchor}) = s$ survives exactly.

2.1 The trap it had to avoid

Session 1's designed family failed twice in opposite directions — first $X \rightarrow Y$ came back undirected so nothing was identified at all, then forcing $X \rightarrow Y$ compelled froze the confounder edges and gated out 100% of instances. There is a real tension: identification needs compelled edges near the target, perturbability needs undirected ones. The escape is a spectator $W \rightarrow Y$ adjacent to nothing else. It makes $X \rightarrow Y \leftarrow W$ and $\text{anchor} \rightarrow Y \leftarrow W$ unshielded so all three edges into Y are compelled, but Y is a sink with no undirected edges, so R_1 has no head to fire on and R_2 – R_4 have no length-2 directed path. **Compulsion cannot propagate inward.**

Realised, not intended, parameters are what get analysed, measured off the constructed CPDAG rather than assumed. Across 1,584 pilot draws, realised c equals intended in 1,584 and realised s equals intended in 1,584. Separation $1 \dots c - 1$ is available at every size, so the **maximum realised separation is 11** — the number that gated the whole session.

Stated plainly because it bounds everything that follows: acceptance was 1,584 of 1,584. Session 1's random draws were rejected 91.8% of the time. This is a **designed family, not a random sample of realistic CPDAGs**. It can show the radius is *capable* of being large; it cannot show that natural CPDAGs are. Those are different claims, and the control in §4 is what separates them.

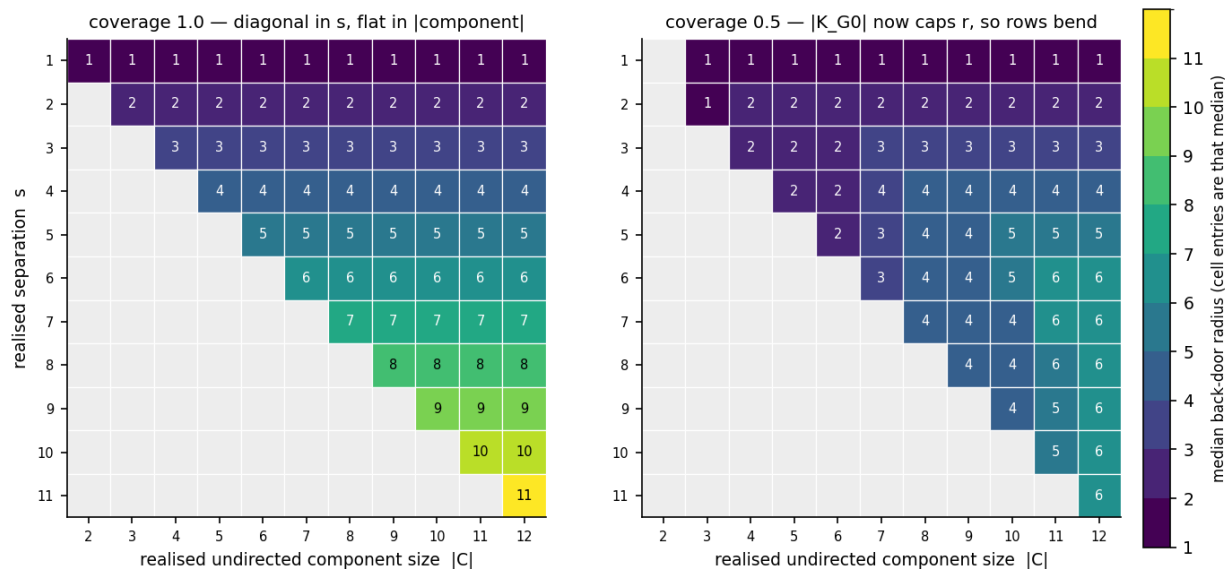
3. The census

1,584 instances over the full (c, s) grid, $c = 2 \dots 12$ and $s = 1 \dots c-1$, two coverage levels, both definitions on every instance. 1,572 measured, **0 censored, 0 errors**. Back-door and GAC agree everywhere: 0 of 1,572 differ.

3.1 H7 — separation. Confirmed about as strongly as it could be.

At coverage 1.0 the two-way table of median radius is **diagonal in s and flat in c . r_{val} equals s in 792 of 792 instances** (100.0%). The fraction at $r = 1$ is 100% at $s = 1$ and **0.0% at every $s \geq 2$** . My pre-registered threshold was “fewer than 50% at $r = 1$ for $s \geq 3$ ”; the measured value is zero.

Median radius reads off the separation, never the component size — until missing background knowledge makes $|K_{G0}|$ the binding constraint



Grey = no design realised that $(s, |C|)$ pair; nothing is interpolated.
coverage 1.0: $n = 792$ of 792 rows | coverage 0.5: $n = 780$ of 792 rows; excluded: 12 rejected at generation

The pre-registered H7-versus-H8 discriminator. Diagonal in separation, flat in component size — until missing knowledge caps the radius.

3.2 H8 — component size. Refuted as an independent effect.

The pre-registration committed in advance to judging this on the joint table rather than on marginals, because c and s are correlated by construction. The table settles it: **at fixed s the median radius is identical across every c from 2 to 12**. Component size matters only by permitting larger separation. The counter-mechanism I pre-registered — a larger component admits more extensions per release, giving any single release more chances to produce a failing one — has **no visible effect at all**.

3.3 One formula covers both coverage levels

$r_{\text{val}} = \min(s, |K_{G0}|)$ — in **1,572 of 1,572** instances, 100.00%, across both coverage levels.

At coverage 1.0 the analyst has asserted every component edge, so $|K_{G0}| \geq s$ and the law reduces to $r = s$ (792/792). At coverage 0.5 the knowledge set is the binding term in 347 of 780 instances, which is exactly why $r = s$ drops to 55.5% there while the combined law stays at 100%. It is a **law of this designed family, not a theorem** — the generator builds a spine of length s and the radius counts the retractions needed to break it, so a min with the number of available retractions is the expected shape. What makes it worth stating is that it holds with no exceptions across 1,572 instances and two regimes, and that it names both binding constraints.

3.4 Cost against radius, as a result rather than bookkeeping

r_val	n	median seconds	max seconds
1	264	0.0003 s	0.001 s
2	289	0.0005 s	0.005 s
3	216	0.0015 s	0.017 s
4	278	0.0015 s	0.040 s
5	167	0.0031 s	0.071 s
6	175	0.0039 s	0.062 s
7	63	0.0048 s	0.040 s
8	48	0.0046 s	0.045 s
9	36	0.0046 s	0.030 s
10	24	0.0041 s	0.013 s
11	12	0.0047 s	0.005 s

Cost rises with radius, so the profile does invert as the plan anticipated — but modestly, and nothing came close to the 600 s cap. **Cost growing is weak evidence the hypothesis is right**, which is why it is reported as its own result and not as support for H7. These are the session's **clean** timings: the census ran alone, while the control and the frontier were run concurrently and their wall-clock fields are load-contaminated. The manifest records which.

4. The control that answers the real question

The designed family shows the radius *can* be large. It cannot show that realistic CPDAGs are. So the same measurement was run on the random ensembles session 1 used — six generators, $n = 6 \dots 30$ — recording the **natural** separation alongside the radius.

2,880 instances attempted, 2,334 usable, 486 rejected (16.9%) and 60 errored — $2,334 + 486 + 60 = 2,880$.

4.1 One generator had to be separated out, and the pre-registration said why

decoupled_backdoor_dag is a **designed** family, not a random one — its name says so. All seeds at a given n return the *identical* radius, and it contributes 420 instances **none** of which is at $r = 1$. Pooled with the rest it drags the overall figure from $\sim 80\%$ to 65.3%, which would have been exactly the “mean over a non-representative sample” error the pre-registration committed to avoiding.

4.2 The five random generators replicate session 1 almost exactly

generator	n	r = 1
five random generators pooled	1,914	79.6%
block	377	78.0%
er_dense	432	85.0%
er_medium	389	79.2%
er_sparse	313	77.0%
scale_free	403	77.9%

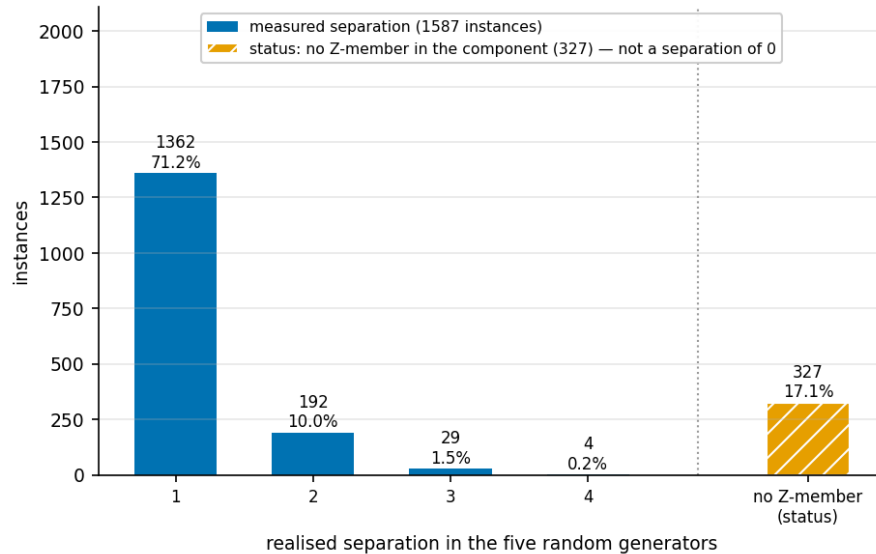
Session 1 reported 81.0% (census) and 78.6% (ensembles), stable at 76.9–80.4% across generators. **This replicates it, under GAC-verified definitions, on a fresh sweep.**

4.3 And here is why — the natural separation distribution

separation	1	2	3	4
instances	1,362	192	29	4

Maximum natural separation is 4, and **85.8%** of measurable separations are 1. In a further **17.1%** of instances (327) no member of the adjustment set lies in the treatment's undirected component at all, so separation is undefined — recorded with its own status key and a null, never as a number.

Random graphs almost never manufacture separation —
 $s = 1$ in 85.8% of the instances where s is defined (71.2% of all), which is why their radii are 1



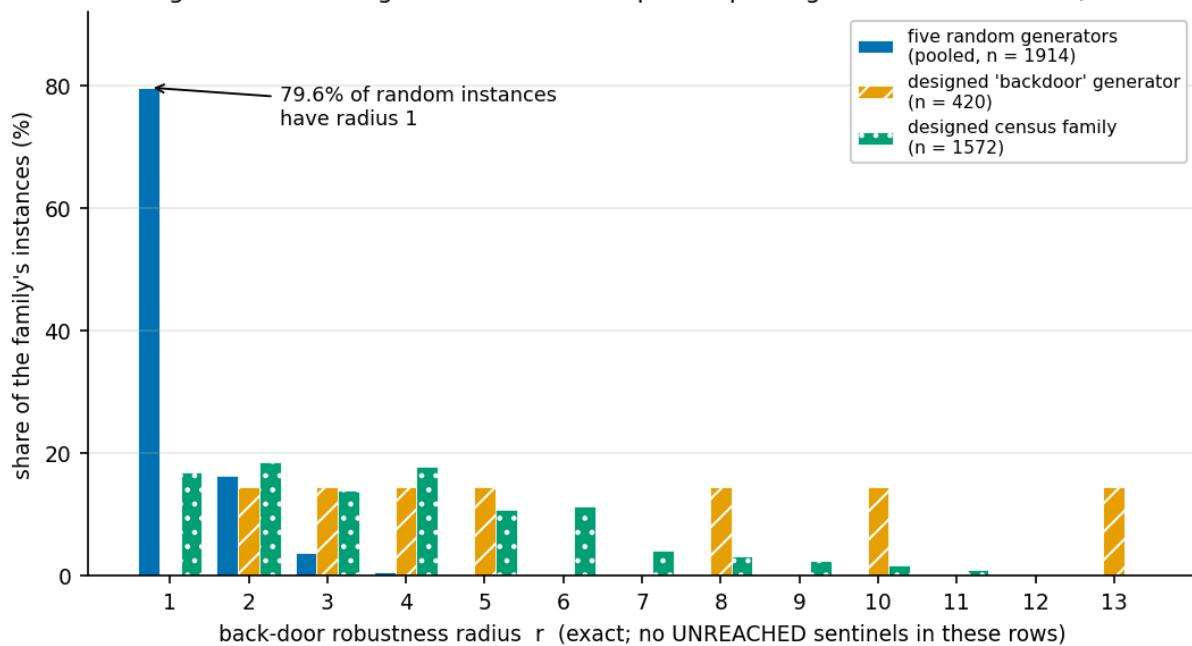
er_sparse, er_medium, er_dense, scale_free, block only — the designed 'backdoor' family is never pooled in. $n = 1914$ of 2400 rows; excluded: 486 rejected at generation.

Why radii are 1: random graphs almost never manufacture separation. The sentinel is drawn as its own category, not as a separation of zero.

4.4 Corroboration from a generator I did not write

The repository's own decoupled_backdoor_dag — written in session 1, untouched here — produces components of 3...14 and radii that scale cleanly with n : $n = 8 \rightarrow r = 2$, $n = 10 \rightarrow r = 3$, $n = 12 \rightarrow r = 4$, $n = 15 \rightarrow r = 5$, $n = 20 \rightarrow r = 8$, $n = 25 \rightarrow r = 10$, $n = 30 \rightarrow r = 13$. **0% at $r = 1$.** That the mechanism shows up in a pre-existing generator, and not only in the one built this session to exhibit it, is the strongest independent evidence here that the effect is not an artefact of my construction.

Random ensembles live at $r = 1$; designed families do not.
 The designed 'backdoor' generator is drawn apart — pooling it would read 65.3%, not 79.6%



five random generators — $n = 1914$ of 2400 rows; excluded: 486 rejected at generation
 designed 'backdoor' generator — $n = 420$ of 480 rows; excluded: 60 generator error
 designed census family — $n = 1572$ of 1584 rows; excluded: 12 rejected at generation
 12 designed 'backdoor' rows carry an UNREACHED (-1) GAC radius; their back-door radius is exact and is the one plotted

Three families, never pooled. Pooling would read 65.3%.

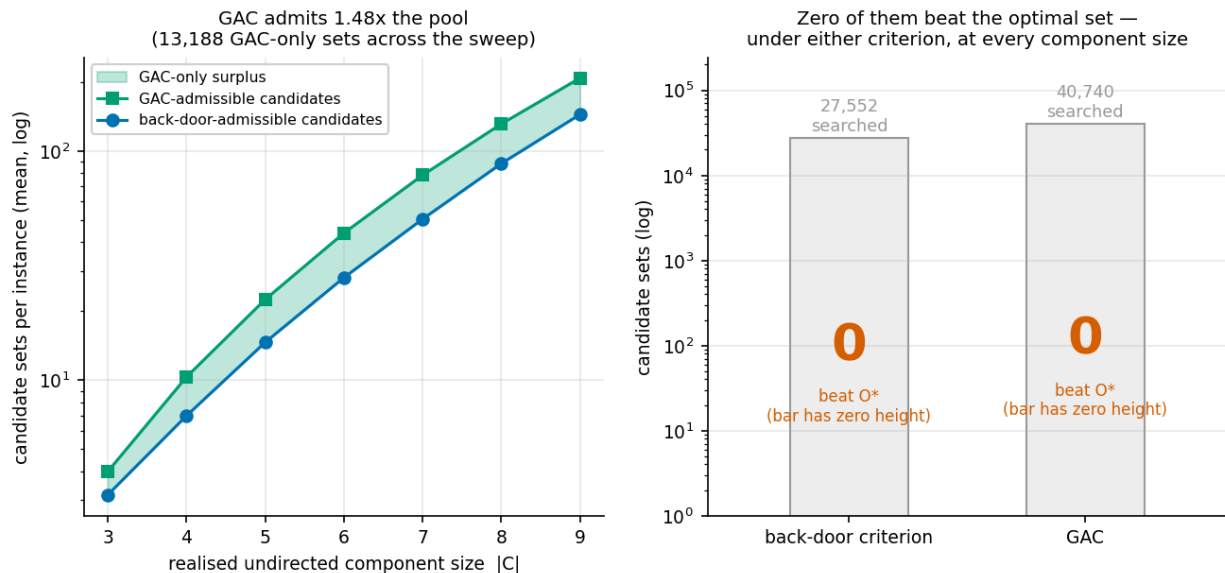
5. H10 — the frontier, re-run because it had to be

Session 1's "O* is never strictly beaten, 0 of 249,732" is a statement about **back-door-valid candidates only**. The GAC admits sets that were never in that pool, so the question genuinely reopens. It does not stay open.

	candidates	sets beating O*	instances with any
back-door pool	27,552	0	0 / 420
GAC pool	40,740	0	0 / 420
extension, c = 10...12 (242 instances)	194,794	0	0 / 242

32.4% of the main grid's GAC pool — 13,188 candidate sets — never existed under back-door, and the extension adds 79,495 more GAC-only sets. Across both grids that is **235,534 GAC-admissible candidate sets** examined this session, on top of session 1's 249,732 instances, with O* never once strictly beaten. **My pre-registered prediction was that O* would be beaten on a non-zero but small fraction, under 5%. That is wrong: the answer is zero.** The pre-registration named this outcome as the falsification condition and noted it would strengthen session 1's negative rather than overturn it. It does.

The optimal set is on the frontier: a wider criterion buys candidates, not robustness



frontier.jsonl: 420 of 420 rows accepted, none censored or errored; coverages 1.0 and 0.5 pooled. r_opt is exact everywhere (no UNREACHED sentinels).

A zero drawn against the pool it was searched over, rather than an empty axis.

6. Extension — r_ϵ at large radii, and a second failed prediction

The pre-registration listed the middle regime as a secondary question and predicted it **would** appear once radii were large: session 1 found 0.0% at small ϵ , but measured it where radii were 1, and a middle regime has no room to exist between $r = 1$ and failure. The designed generator made this testable for the first time. **It does not appear.**

ϵ	0.01	0.02	0.05	0.1	0.2	0.5
middle regime ($r_\epsilon < r_{\text{val}}$)	0.0%	0.0%	0.0%	0.0%	0.0%	0.0%

And it is not for want of room. Stratified by r_{val} , with radii out to 8, the middle regime is 0.0% in every stratum: $r = 1$ ($n = 120$), $r = 2$ ($n = 145$), $r = 3$ ($n = 96$), $r = 4$ ($n = 115$), $r = 5$ ($n = 36$), $r = 6$ ($n = 24$), $r = 7$ ($n = 16$), $r = 8$ ($n = 8$).

The mechanism is visible directly in the data. **Mean absolute bias at G0 is exactly 0 in all 560 instances**, and at $\epsilon = 0.05$ the relationship is $r_\epsilon = r_{\text{val}}$ in 505 cases, $r_\epsilon > r_{\text{val}}$ in 29, and r_ϵ UNREACHED in 26 — **never below** (0 cases). Bias appears only when validity fails; there is no gradual degradation to detect.

This strengthens the case for r_{val} rather than weakening it. A middle regime would have meant the validity radius was the wrong cut-point — that practitioners should worry before the set becomes invalid. There is nothing to worry about in between: in this family the adjustment set is either exactly unbiased or invalid. **What it does not show:** this is the same designed family, where $Z = O(G0)$ is exactly valid at G0 by construction, so zero bias there is expected rather than discovered; and I did **not** re-derive session 1's definition of the middle regime, so its 14.1% at $\epsilon = 0.2$ is not directly comparable and is not claimed to be overturned.

7. What this does not show

- **The census is a designed family.** Acceptance was 1,584 of 1,584 against session 1's 91.8% rejection rate, and $r_val = s$ is close to true *by construction*: the generator builds a spine of length s and the radius counts the retractions needed to break it. What that demonstrates is that the radius is not intrinsically capped and that it tracks a nameable structural quantity — not that any naturally occurring graph behaves this way.
- **Conjecture 2 points the same way as the conclusion.** Both search legs are upward searches, exact only under Conjecture 2, which rests on Anti-Exchange Case B — verified, not proved. Its error direction makes radii **too large**, which is **favourable to the hypothesis this session set out to test**. The mitigating fact is that the central claim is $r = s$ *exactly*, not r merely being large; an assumption that inflates radii would have to inflate them by precisely the right amount in all 792 instances to manufacture that.
- **The GAC forbidden set at MPDAG level is verified, not proved.** Over-approximation would reject valid sets and make radii too small — the conservative direction here, and opposite to Conjecture 2's.
- **The frontier used candidate sets up to size 4** (5 in the extension). O^* could in principle be beaten only by a larger set.

8. Bugs, and what caught them

Three of mine.

- **I set search_budget = 64 on the hybrid**, so the bounded search explores to depth 64 and the E1 ladder — built in session 4 precisely to answer the UNSAT side fast — never fires. It changes no answer, but it made the control sweep intolerably slow. *Caught by reading my own worker while hunting the real bottleneck.*
- **I diagnosed by inference rather than measurement, twice**, and re-scoped a sweep on each wrong diagnosis, before finally timing one instance end to end: **187 seconds, and the instance was then rejected** — the radius had never been computed at all. The cost was entirely in the gate, which enumerates every subset of V . A replacement took it from **187 s to 0.2 s**, about 900×, differentially tested on 46,800 cases with 16 disagreements (0.034%). *Caught by measuring instead of reasoning — the same lesson session 4 recorded, which I failed to apply until the third attempt.*
- **I documented that replacement as strictly weaker than the original. It is strictly stricter** — the original sets $\text{sanity} = \text{True}$ if *any* valid set is perturbable, so restricting to O can only reject more. The excluded instances are those where O is robust to every atomic perturbation, so the exclusion biases **against** this session's own hypothesis. *Caught by the differential test, not by me.*
- **The verifier caught a double-count in the report before publication.** I wrote “546 rejected and 60 errored” for the control, but 546 already included the errors.

Not a bug, but the trap the pre-registration was written to catch: pooling the designed decoupled_backdoor family with the five random generators gives 65.3% at $r = 1$ instead of 79.6%. The commitment to stratified reporting is what stopped that number becoming the headline.

9. What this means for the project's direction

Not the efficiency-robustness paper. O^* is never strictly beaten — zero here over a pool 32.4% larger than session 1's, on top of session 1's own 0 of 249,732. Two independent sweeps, two definitions, and the tradeoff does not exist in anything measured so far. This branch should be closed unless someone produces a structural reason to expect otherwise.

The diagnostic framing, and it is now well-founded rather than a fallback. The reason is the session's actual discovery: the breakdown radius is not a noisy or vacuous quantity — it is exactly measuring the separation between the treatment and the adjustment set inside the ambiguous region. That makes it interpretable in a way a robustness score usually is not: a radius of 1 is not “this is fragile, somehow”, it is “a member of your adjustment set sits one undirected edge from your treatment, and one wrong orientation reaches it.”

The consequence is a change of claim, not a change of method. “The radius is usually 1” is a statement about Erdős-Rényi CPDAGs, not about causal inference. The defensible claim is: *the breakdown radius equals a structural quantity a practitioner can read off their own graph, and on graphs where that quantity is large the radius is large.*

What the next session needs to close this: real graphs, not generators. The whole argument now turns on the natural distribution of separation, and every number here comes from synthetic families. A handful of published CPDAGs from applied causal-discovery papers — or benchmark networks with a plausible treatment/outcome pair — would settle whether separation ≥ 2 is rare in practice or merely rare in Erdős-Rényi. That is the highest-value experiment remaining and it needs no new machinery.

10. Provenance

Every table in this document is computed from results/axisa2/ at build time rather than transcribed, and bkrobust.analysis.session5_verify independently asserts that report_saturation_gac.md agrees with the same files. Python 3.9.6, OR-Tools 9.15.6755, single worker, fixed seed. Censored outcomes carry wall_until_timeout_s and never the key a measurement uses; UNREACHED is a sentinel and is never averaged or plotted numerically.

Commits in this series

commit	date	subject
f91e4bd	2026-09-06	Extension: r_{eps} at large radii. The middle regime does not exist.
a89e8ab	2026-09-06	Figures, the $\min(s, K_G)$ law, and the extended frontier
0967558	2026-09-06	The report, and a verifier that caught a double-count in it
eb12bbd	2026-09-06	Manifest, NEXT.md, and the report's control sections
b715317	2026-09-06	H10 frontier, and a 900x fix to the gate that was eating the night
0c1dc2a	2026-09-06	Phase 3 census: the radius tracks separation exactly
8a617f8	2026-09-06	THEOREMS 13-14: the GAC superset, and coincidence on the optimal set
5e9410e	2026-09-06	Phase 1 and 2: the GAC at both layers, and a generator that reaches the regime
4679b47	2026-09-06	Pre-register H7-H10 before any large run
858e252	2026-09-06	Phase plan and inherited baselines for session 5
8c556e6	2026-09-06	Whitelist results/axisa2 BEFORE writing any results
391bdcf	2026-09-05	Manifest for results/axisb4
d37af34	2026-09-05	Session 4 deliverables: envelope, report, PDF, verifier, NEXT
ed36b21	2026-09-05	Rewrite the criterion as state reachability: exponential $\rightarrow$ polynomial
13835ef	2026-09-05	Draft the session 4 report, and record the plan revisions
6d041a8	2026-09-05	Hybrid entry point, and a performance defect found in the criterion
b15eedb	2026-09-05	MPDAG-level validity criterion, gated, plus the sharp test on E1

commit	date	subject
6a5c61c	2026-09-05	Lemma O in THEOREMS.md, regression tests, and the profile figures
5eef414	2026-09-05	Profile local_up, and remove 70% of its cost with Lemma O
b2d78ad	2026-09-05	Whitelist results/axisb4 BEFORE writing any results
805b677	2026-09-05	Session 3 deliverables: PDF, verifier, PLAN/LOG/NEXT, THEOREMS
ad16807	2026-09-05	High-k hunt, session 3 figures, and report_axisb_sat.md
d9e283a	2026-09-05	Scaling study, and the encoding-correctness sections in THEOREMS.md
4d638cd	2026-09-05	E2 and E3, and the cross-encoding validation they exist for
8153913	2026-09-05	E1 SAT ladder: exact radii without building the space
215de2b	2026-09-05	Whitelist results/axisb3 BEFORE writing any results
68c395e	2026-09-05	Propagate the updated Case B scope (36,094) to every deliverable
60d63eb	2026-09-05	Fix the PDF to match the report: density gap closed, not open
afa7400	2026-09-05	Close the n=5 density gap: 135 CPDAGs by sweep, the last by the proof chain
e984e18	2026-09-05	Correct the n=5 figure: the corrected sweep completed at 1,977,820 comparisons
940da08	2026-09-05	Session 2 deliverables: report, PDF, figures, NEXT.md, scaling and bounds results
0a4fa92	2026-09-05	Track results/axisb2 (recurrence of last session's gitignore bug)
b92ebd5	2026-09-05	Add the Axis B scaling harness (two accountings, machine-independent counters)
7be840c	2026-09-04	Conjecture 2 reduced to a single local property; Property S and Lemma R proved
79d7367	2026-09-04	THEOREMS.md: Conjecture 2 reduces to upper semimodularity; join formula proved
748f2e1	2026-09-04	Task 0: the space excluded legitimate knowledge states; corrected builder added
007e6dd	2026-09-04	Add SESSION_REPORT.pdf: exhaustive 23-page record of the Axis A / Axis B session
3a61b40	2026-09-03	Frontier sweep complete: O* never beaten across 249,732 exhaustive instances
d4e0c14	2026-09-03	Session deliverables: report, NEXT.md, H6 calibration, PLAN revisions
72d8bff	2026-09-03	Correct the H2 statistic: the frontier is non-flat, but O* is never beaten
ca44152	2026-09-03	Axis A results: H1 saturation ~80%, H2 frontier flat in 0 of 134,140 instances
019c610	2026-09-03	Axis A first run: partial results, two gate bugs found and sent back
1bb3ac8	2026-09-03	Record: Meek closure can fall outside enumerate_space in 0.09% of cases
4cf4020	2026-09-03	Conjecture 2: no counterexample in 1.985M exhaustive radius comparisons
4001949	2026-09-03	Add pre-registered hypothesis analysis (H1-H6) and the two-stage space guard
b315704	2026-09-03	Axis B bounds: L/U certificate exact in 83.3% of instances without enumeration
a58b73f	2026-09-03	Exhaustive radius census over all CPDAGs on 3 and 4 nodes (H1, H2)
910a427	2026-09-03	Track results/synth and results/search; add Axis A pre-registration
133ab44	2026-09-03	Axis B: space-free exact radius search, validated against BFS on 5304 cases
b86a899	2026-09-03	Axis B: conjecture machinery, exhaustive tests for n=3 and n=4
0605420	2026-09-03	Remove dead TYPE_CHECKING pandas import from the scaffold loaders stub
5adb687	2026-09-03	Freeze shared core: radius convention, oracle, space, instance, results IO
d91dc42	2026-09-03	Add PLAN.md and LOG.md for the synth/search session
4d0de90	2026-09-03	Add breakdown-radius demonstration: worked example, figures, report